

Classifying Kepler Objects of Interest: Confirmed Planets, Candidates, and False Positives

Problem

The Kepler space telescope flagged thousands of periodic dimming events as possible planets. Each Kepler Object of Interest (KOI) eventually gets dispositioned as CONFIRMED (a real planet), FALSE POSITIVE (usually an eclipsing binary star or an instrumental artifact), or stays a CANDIDATE. This project builds a classifier that predicts the disposition from the NASA Exoplanet Archive cumulative KOI table (9,564 objects, 140 catalog columns) and explains, in physical terms, what drives its decisions.

Data handling and EDA

The extract deliberately excludes `koi_score`. That column is a direct output of the vetting process that produced the label, so using it would be target leakage, and we made no attempt to reconstruct it. EDA surfaced three structural issues.

Nineteen columns are entirely null (eccentricity errors, stellar age, model fit statistics, and similar). They carry no information and were dropped.

Identifier and leakage-prone columns were removed from the feature set: `kepid` and `kepoi_name` are IDs, `koi_pdisposition` is a near duplicate of the target, and `kepler_name` is only assigned to confirmed planets, so even its missingness encodes the answer.

KOI uncertainties come in symmetric $\pm$ pairs. A correlation check confirmed that most `_err2` columns are perfectly anti-correlated with their `_err1` twin, so the mirrors were pruned.

Remaining missingness (at most 16% per column, concentrated in secondary vetting statistics) was median-imputed, with medians fit on the training split only. Most physical quantities here are heavily right-skewed, and a mean would be dominated by the tails. The few low-cardinality categorical columns (fit type, catalog provenance) were one-hot encoded with "missing" as an explicit level.

Class balance is 51% FALSE POSITIVE, 29% CONFIRMED, 21% CANDIDATE, so we optimized macro F1 rather than accuracy and used balanced class weighting in every model. We preferred weighting to synthetic oversampling (SMOTE) because interpolating in a feature space of heavy-tailed physical quantities can fabricate physically impossible objects. Weighting keeps every real observation.

Feature engineering

Eight domain-derived features were added, each tied to a piece of vetting physics. Transit depth per hour separates stellar eclipses, which are deep and sharp, from planetary transits, which are shallow and gradual. The planet-to-star radius ratio catches "planets" comparable in size to their host star, which are in fact stars. The ratio of the multiple-event to the single-event statistic captures whether evidence accumulates over repeated orbits, as it does for real planets, or arrives once, as it does for glitches. Comparing the observed transit duration against the value expected from Kepler's third law flags grazing eclipses. Log transforms of period, depth, and insolation handle quantities that vary over orders of magnitude. Finally, SNR per transit separates intrinsically strong signals from ones that merely got observed many times.

Models and results

Three models of increasing capacity were compared with stratified 5-fold cross-validation on an 80/20 stratified split, with the random seed fixed at 42 throughout:

Model	Test accuracy	Test macro F1
Logistic Regression (baseline)	0.869	0.776
Random Forest	0.884	0.821
XGBoost	0.859	0.830

A 20-configuration randomized hyperparameter search on XGBoost moved cross-validation macro F1 only slightly and did not improve the held-out test score. The defaults were already near optimal, and we report that as is rather than cherry-picking a run. Per-class F1 for the best model: FALSE POSITIVE 0.907, CONFIRMED 0.903, CANDIDATE 0.682. Nearly all the residual error in the confusion matrix sits on the boundary between CONFIRMED and CANDIDATE, which makes sense: a candidate is, by definition, a planet-like signal that has not yet cleared the confirmation bar, so the two classes genuinely overlap.

One decision deserves its own paragraph. The four `koi_fpflag_*` columns are outputs of Kepler's automated vetting pipeline — they encode the very false-positive classification we are trying to predict. Including them as features is textbook data leakage. We therefore exclude them entirely from the feature set. The model is trained on transit-physics and stellar-parameter features alone, and still achieves strong results, with most error concentrated on the CONFIRMED/CANDIDATE boundary where it belongs.

What the model learned

SHAP analysis shows that the model relies on genuinely physical features: the multiple-event statistic (is the signal periodic and cumulative?), the centroid offset `koi_dicco_msky` (is the dimming actually happening on the target star, or on a neighbor blended into the same pixel?), transit signal-to-noise (confirmation requires a clean signal, and low SNR is a big part of why candidates stay candidates), and planet radius (objects larger than about 2 Jupiter radii are almost always small stars). This decision logic mirrors how astronomers vet KOIs by hand: check the geometry, check the repeatability, check the signal quality. That is decent evidence the model learned physics rather than dataset quirks.

Explaining the predictions to a non-technical audience

Imagine Kepler as a security camera pointed at a hundred thousand stars, watching for a light in the house to flicker. Most flickers turn out to be nothing: a passing car, a bug on the lens. A real burglar walks past the same window on every lap, dims the light by the same small amount each time, for the same number of seconds. That regular, faint, repeating dimming is what a real planet looks like, and it is what the model is trained to recognize.

The model checks a short list of things an astronomer would check by hand. Does the dimming happen on the actual star, or on a neighboring star caught in the same pixel? Does it repeat on schedule, or did it happen once and never again? Is the dip shallow and gradual, like a small planet sliding in front of a much bigger star, or deep and sudden, like two stars of similar size eclipsing each other?

Where the model struggles is the same place a human would struggle: telling a confirmed planet from a candidate. A candidate is not a different kind of object, it is a planet-shaped signal that has not yet accumulated enough evidence to be certain. So when the model mixes up these two categories, it isn't making an error so much as reporting genuine uncertainty, in the same way a doctor might say "probably benign, pending more tests" rather than giving a false yes or no.

Reproducibility

Everything runs top-to-bottom from two notebooks (`01_eda.ipynb`, `02_modeling.ipynb`) backed by four documented modules in `src/`, with fixed random seeds and pinned dependency ranges in `requirements.txt`.